

Power, Participation, and Knowledge Production: Technology Mediated Epistemic (In)Justice

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Abstract

Epistemic injustice is the systemic exclusion of certain people and methods from knowledge-producing processes. Building on my larger arc of studying how justice manifests in new technologies, my dissertation studies technology-mediated epistemic injustice—how our theories and policies of online knowledge are often exclusionary and unjust. Despite naming technology as the problem, I also envision it as the solution. I start by identifying contributing factors to epistemic injustice in online settings. Based on these findings, I propose structural changes to research processes that propagate epistemic injustice. Additionally, I propose a design solution: a Socratic LLM for navigating Wikipedia’s dense and socially constructed epistemology.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms.**

Keywords

injustice, epistemology, online communities, wikipedia

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1 Introduction

Epistemic injustice is when one is distanced from their due right to engage in knowledge-oriented process [9]. Like most forms of injustice, epistemic injustice is animated by power structures, prejudice, and exclusion. Indeed, to distance someone from their due credibility and sensemaking capacity is to fundamentally disempower them.

While epistemic injustice may seem abstract, it compounds concrete social harms that can cost lives. Offline, epistemic injustice manifests in numerous settings, such as women’s testimony of their pain not being believed by doctors [7], and sexual assault survivors

not having the vocabulary to describe their experience [11]. Online, epistemic injustice often manifests as disparities in codified information and access. For example, only 20% of biographies on Wikipedia are about women—signaling the larger gender gap on Wikipedia [17]. For a higher-stakes example, previous CSCW work describes the disproportionate deplatforming of trans women, particularly those sharing information on gender-affirming care [8]. This systemic lack of online health knowledge and community contributes to unconscionable suicide rates in trans populations [14]. If these forms of technology-mediated epistemic injustice are left unresolved, we risk never truly serving marginalized users.

My research area lies at the intersection of knowledge formation, justice ethics, and technology. Using epistemic injustice as my motivating theory, I seek to understand how key technologies in CSCW—such as online communities and predictive systems—both propagate and create forms of knowledge-based marginalization. How do the unique rules, norms, and politics of online communities affect knowledge? In addition to problematizing epistemic injustice in CSCW, I suggest that research and design can be a powerful tool in the fight against epistemic injustice. Building on work that uses Wikipedia as a formative example of knowledge politics [15], I ultimately want to build an effective intervention that challenges the epistemic status quo. Given these goals, my dissertation will answer the following research questions:

- RQ 1. How does epistemic injustice manifest in HCI venues?
- RQ 2. What are the current contributors to epistemic injustice in these venues?
- RQ 3. How can we design technological interventions to aid users in reflecting on their epistemology?

2 Preliminary Work

To address research questions 1 & 2, I have completed theoretical work ([2], [5]) on stories of epistemic injustice in common HCI venues, such as peer-production communities, social media, and even research itself. I employed collaborative autoethnographic methods. I found that this method was helpful at capturing both the breadth of epistemic injustice venues and the distinctly personal experience of epistemic injustice. While I exclusively collaborated with CSCW researchers for [2], I expanded the method to incorporate activists with no prior research experience as autoethnographers [5].

The findings from these two studies have allowed me to articulate the nuances of technology-mediated epistemic injustice. First, I found that technology reifies offline epistemic injustice. One participant described the current state of online trans health information; articles written by an organization of trans doctors,

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TPATH¹, were consistently moderated off of platforms as non-credible sources. As such, two forms of epistemic injustice surface: (1) trans folks are further distanced from potentially life-altering knowledge around gender-affirming care, and (2) trans doctors creating this community-oriented knowledge are undermined as medical authorities. The lack of health access and credibility that trans folks experience offline is being replicated and compounded online.

Second, technology creates new forms of epistemic injustice. Participants described various identity-based online communities they are a part of, such as the subreddit about bisexuality (r/bisexual) and mental health-related corners of TikTok. As a way of interpreting their lived experience, these communities often create their own hermeneutic resources that capture the imprecision of identity. For example, one participant explained the term “neurospicy” as a colloquial term on TikTok to capture the wide spectrum of neurodevelopmental differences. Despite this powerful term to interpret a marginalized experience, it became a target for auto-moderators. This removal creates a form of epistemic injustice where users are stripped of their ability to understand their own mental health experience and, subsequently, build community. Essentially, the moderator algorithm weaponized the community’s hermeneutic resource.

Finally, both research and design are unique paths to fighting epistemic injustice. Research itself is an epistemic community: a group of people dedicated to forming, producing, and disseminating knowledge. Yet, I found that many research practices distance our participants from the research that we publish about them. I articulate this as a form of epistemic injustice and propose the antidote is *epistemic autonomy* [5]. Epistemic autonomy is one’s ability to govern knowledge about oneself, derived from the intersection of epistemic injustice and trans feminist autonomy. I posit that epistemic autonomy is a mechanism for moving closer to justice within our research practices. I suggest that achieving epistemic autonomy involves both philosophical changes, such as accepting that knowledge is socially constructed, and methodological changes, such as inviting participants in as research collaborators where appropriate. Throughout my own research, I hope to exemplify these changes.

3 Work in Progress

In my current work, I plan to move from understanding technology-mediated epistemic injustice (RQ1, RQ2) and toward actively remediating it (RQ3). I plan to use Wikipedia as a test bed for designing a tool that remediates epistemic injustice within a specific online community. Achieving any form of justice requires a mixture of big systemic changes and smaller moments of individual reflection. Above, in works [2, 5], I address the systemic changes researchers and designers can facilitate. Below, I describe two works in progress focused on the latter.

3.1 Becoming a Knower on Wikipedia

To start, I investigated current contributors to epistemic injustice on Wikipedia. To orient towards a design outcome, this work focuses on Wikipedians’ interactions with the platform’s knowledge-oriented policies and practices. Despite this interaction focus, this

research question is in conversation with previous literature on the platform’s structural inequities [15, 17]. I invited Wikipedia activists, scholars, and Wikimedia Foundation employees to a two-day design workshop. In this workshop, I integrated the collaborative autoethnography methods from my previous work with co-design activities. This mixture of reflective and generative activities allowed us to produce (1) an expert-driven articulation of epistemic injustice on Wikipedia and (2) expert-designed prototypes of possible Wikipedia redesigns. In the spirit of epistemic autonomy, all workshop participants were invited into the paper writing process and are authors on the current manuscript.

Our workshop participants identified how gaining influence over Wikipedia’s policies is extremely difficult. This epistemic burden disproportionately affects communities with marginalized histories or alternate ways of knowing. For example, one participant noted that her local queer community was forced underground in the 1970s and, therefore, their history is documented in grassroots zines, newsletters, and oral histories from elders. These non-traditional sources often fall on the wrong side of Wikipedia’s reliable sourcing policies, thereby limiting the amount of this history that is allowed on the platform. Compounding this exclusion, Wikipedia policy is dense, difficult to bend, and even more difficult to change. Therefore, there is little recourse for communities whose knowledge is not traditionally documented. Our participants synthesized the solution into a three-step process of easing the following processes: *policy learning*, *interpretation*, and *renegotiation*. Calling back to the epistemic injustice literature [9], we call this process “*becoming a knower*” on Wikipedia. Each of these three steps has unique implications for re-designing the platform. Participants designed simple prototypes, like auto-suggesting relevant policies while a user is writing an edit, and radical ones, like forking Wikipedia into numerous communities, each with their own epistemology.

3.2 Socrapedia: Reflecting on an Epistemology

Based on my prior work, we now have (1) a robust concept of technology-mediated injustice and (2) an expert-driven understanding of contributing factors. Chat-based LLMs were raised in the Wikipedia workshop as they offer a powerful tool for considering a community’s policies and contextually relaying them to a user. Specifically, LLM-powered chatbots can facilitate dialogic methods, such as the Socratic method [6]. In philosophy, the Socratic method is considered a meta-epistemic method—proven to help an individual understand knowledge-oriented processes. The Socratic method involves one person (i.e., “Socrates”) asking a student a series of interrogative questions that prompt the student to question their own beliefs. I want to test whether introducing a Socratic LLM dialogue system can facilitate Wikipedia policy learning and interpretation. In other words, **can a Socratic LLM help Wikipedia editors become knowers?** I am currently developing a Socratic LLM that uses Retrieval-Augmented Generation (RAG) to tune the model on Wikipedia’s policy documents. The resulting model will be Socrapedia: an LLM prompted to engage in the Socratic method and tuned to Wikipedia’s epistemology.

To test Socrapedia, I plan on running a user study with both Wikipedians and non-Wikipedians. In the treatment condition, my study would take participants through three parts: (1) introducing

¹<https://tpathealth.org/>

a disputed Wikipedia edit, (2) requiring the participant to engage with our Socratic LLM, and then (3) asking them how they would resolve the dispute. The control condition would omit Step 2. To seed these tasks, I will use a dataset of controversial Wikipedia edits and outcomes from my previous work [1]. This dataset provides further control data as it traces the actual outcomes that occurred on Wikipedia. Further, I plan to analyze participant outcomes comparatively by their Wikipedia experience level, leading to two hypotheses. First, I hypothesize that participants with minimal or no Wikipedia experience will heavily align with our ground truth dataset. Second, I hypothesize that participants with mid-to-high levels of Wikipedia expertise will significantly deviate from our ground truth. If true, this experiment would validate that our tool facilitates policy bending and could inform future policy changes. While these methods deviate from my qualitative work, they build on my previous work in system design [13] and quasi-causal analysis [4]. In merging these three methodological skills, I hope to create a dissertation that formulates both a problem space and a solution space for epistemic injustice.

4 Anticipated Contributions

- **A translation of epistemic injustice as a lens for understanding CSCW communities.** While I focus on Wikipedia as a test bed, it is far from the only online community with an epistemology. Previous CSCW work has studied similar public scholarship sites, such as r/AskHistorians [10]. Social media sites are creating their own forms of knowledge dissemination, through “algospeak” and community norms [16]. Therefore, I anticipate that my translation of epistemic injustice can unify research agendas across CSCW sites of study.
- **Co-design methodology that embraces epistemic autonomy for participants.** A product of studying epistemic injustice is reflecting on how my own research practices may inadvertently contribute to the problem. As such, I hope the methods presented in my dissertation, such as the autoethnographic design workshop with Wikipedians, provide an example of moving towards epistemic autonomy in HCI research.
- **Designs to help knowledge-formation communities reason about epistemology.** While design solutions must occur in tandem with structural change, they offer a promising path towards remediating epistemic injustice. My dissertation makes headway down this path by presenting co-designed low-fidelity prototypes from Wikipedians and testing a fully built tool inspired by those prototypes. Therefore, my dissertation contributes a design solution to the large problem space of epistemic injustice.

5 Benefits of the Doctoral Consortium

My work both studies canonical CSCW sites, such as Wikipedia, and reflects on the CSCW itself. Therefore, the 2025 Doctoral Consortium will give me the unique opportunity to receive feedback from the CSCW community. At the time of the consortium, I will be in the process of building Socrapedia and designing my user study. Therefore, the consortium will allow me to hypothesize on

the efficacy of Socrapedia, plan for potential study confounds, and test my conceptual argument of remediating epistemic injustice through a meta-epistemic chatbot. I also hope to participate in the consortium by giving feedback to my peers. Given my work has relied on workshops at CSCW [3, 12], I have a deep appreciation for peer support and constructive discussions at this conference.

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